Kirk Long, Ph.D.

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Research Interests

I am broadly excited about employing advances in modern computing to analyze large data-sets and to simulate interesting systems numerically, techniques I am currently using to better understand the true nature of the broad emission line region in active galactic nuclei. I am excited to apply these techniques as a Joint Postdoctoral Fellow at KIAA in Beijing and MPE in Munich starting this fall, where I will use the GRAVITY+ instrument to further probe the dynamics and structure of the broad-line region in AGN across cosmic time.

UAT keywords: active galactic nuclei, supermassive black holes, computational astronomy

Education

08/2020 –	**University of Colorado Boulder**, Dept. of Astrophysical and Planetary Sciences Ph.D. thesis: "Unraveling the Quasar Broad Emission Line Region" (degree conferred May 2026) M.Sc. awarded December 2023: "A Possible Thin Disk-Wind Launching Mechanism of Broad-Line Emission in AGN Applied to Quasar 3C 273"
08/2017 – 05/2020	**Boise State University**, Honors College B.Sc. in Physics, Astrophysics emphasis, *Magna Cum Laude* Minors in Music and Applied Mathematics, recognized as a Graduating Student Leader
08/2015 – 05/2017	**Idaho State University** (transferred prior to completing degree)

Publication List (* indicates student-led projects)

"A census of echo maps recovered from AGN light curves with machine learning,"*****
Williams, P., **Long, K.**, Horne, K., & Dexter, J., 2026, *in prep*

"A Convolutional Neural Network for the Recovery of Transfer Functions From Velocity-Resolved Reverberation Mapping Data," **Long, K.**, Horne, K., Dexter, J., & Tremblay, B., 2026, *ApJ*, 1003, 72

"`BroadLineRegions.jl`: A fast and flexible toolkit for modeling the broad-line region (BLR) in Julia," **Long, K.**, 2025, Submitted to *JOSS* July 2025 (awaiting reviewer)

"Reverberation Mapping Data of NGC 5548 Imply a Multicomponent Broad-line Region," **Long, K.** & Dexter, J., 2025,*ApJ*, 987, 196

"Confronting a Thin Disk-Wind Launching Mechanism of Broad-Line Emission in AGN with GRAVITY Observations of Quasar 3C 273," **Long, K.**, et al. 2023, *ApJ*, 953, 184

Awards

2026	R.N. Thomas Award —recognizing the most outstanding graduate student in astrophysics at JILA
2025	GPSG DEI Excellence Award
2021 – 2022	Astrophysics Graduate Fellowship (APS department prize)
2015 – 2020	Dean's List and $22,000 in undergraduate scholarship awards

Press

2026	"CURC User Success Story—Measuring the masses of black holes and characterizing the environments around them", *CURC Newsletter*
2023-2024	"Questions about Quasars: How to Best Weigh a Celestial Body", *JILA Light & Matter*, same paper also featured in *SciTechDaily*
2019-2020	"Physics Student Brings Science Class to Prison", Boise State University website, local news channel KIVI, and Boise State University alumni magazine, *Focus*

Selected Presentations (*indicates invited)

07/2026	**First Friday Astronomy Public Lecture***, *Boise State University* "How to weigh a supermassive black hole: insights from quasars across cosmic time"	Boise, USA
06/2026	**Supermassive Black Holes and Blue Notes**, *Campus MIL, Université de Montréal* Contributed talk: "Unraveling the broad-line region: novel modeling and deep learning techniques"	Montreal, CA
01/2026	**AAS 247**, *247th meeting of the American Astronomical Society* Contributed dissertation talk: "Unraveling the Quasar Broad-Emission Line Region"	Phoenix, USA
09/2025	**AstroLunch Seminar***, *University of St Andrews School of Physics and Astronomy* "Unraveling the Quasar Broad-Emission Line Region"	St Andrews, UK
09/2025	**Massive Black Holes Across Cosmic Time**, *Kavli Institute for Cosmology, University of Cambridge* Contributed talk: "Unraveling the Quasar Broad-Emission Line Region"	Cambridge, UK
07/2025	**Vasto Accretion Meeting**, *Organized by Durham University and the INAF* Contributed talk: "Unraveling the Quasar Broad-Emission Line Region"	Vasto, IT
06/2024	**SMBH Sexten**, *Organized by Chalmers & Virginia Initiatives on Cosmic Origins* Contributed talk: "Unraveling the Quasar Broad-Emission Line Region"	Sexten/Sesto, IT

Selected Teaching and Research Mentoring Experience

05/2026	**Supervision of undergraduate research student** **Philip Williams**, JILA, University of Colorado Boulder **Project:** Further development and application of novel deep learning techniques for producing echo maps of the broad-line region in active galactic nuclei.
01/2023 – 05/2023	**Graduate Part-Time Instructor** CU Boulder Dept. of Astrophysical and Planetary Sciences Instructor of record for ASTR 2030 Black Holes with enrollment of >100 students. Responsible for all course content and managing TA, grader.
08/2020 – 12/2023	**Graduate Teaching Assistant** (4 semesters)

	CU Boulder Dept. of Astrophysical and Planetary Sciences Taught recitations/labs for both lower-division and upper-division courses, assisted with grading assignments/exams, and occasionally assisted in the development of class materials (like Jupyter notebook labs). **Courses TA'd:** ASTR 3730 (Astrophysics I), ASTR 1040 (Accelerated Intro to Astronomy II), ASTR 2030 (Black Holes)
09/2023 & 10/2025	**Invited Guest Lecturer**, CU Boulder Dept. of Astrophysical and Planetary Sciences Gave lectures on black holes for Prof. Jason Dexter's ASTR 2030 Black Holes and Prof. Erica Nelson's ASTR 1200 Stars and Galaxies courses, each with enrollment of >200 students

Selected Service and Outreach

01/2019 –	**Volunteer Instructor**, Idaho & Colorado Correctional Facilities Taught introductory programming classes, gave physics and astronomy demonstrations/lectures, and helped > 50 students obtain their GEDs.
01/2020 –	**Open-source software contributor** (`@kirklong`) Contributed to various open-source Julia and Python projects as well as distributing my own code in an easily installable format. Top ~2% contributor on StackOverflow. Notable software releases: • `BroadLineRegions.jl` – A fast and flexible toolkit for modeling the broad-line region (BLR) in Julia. • `BinnedStatistics.jl` – An analogue to SciPy stat's `binned_statistic` benefitting from performance of native Julia.
12/2019 –	**@ThreeBodyBot**, #scicomm Built automated Twitter/Mastodon/Tumblr/BlueSky/YouTube account that posts random three-body simulations ~1/day. Source code here.
06/2017 – 08/2020	**Park Astronomer**, Bruneau Sand Dunes State Park Observatory Operated large telescopes to show visitors celestial objects and gave ~45 minute public talks on astronomy topics. Interacted with >20,000 total visitors during employment.
08/2020 –	**Graduate mentor**, CU Boulder (5 students) Mentored both undergraduate and incoming first-year graduate students on both research and academic topics.
08/2020 –	**Graduate committee member**, CU Boulder Served on several department service committees with faculty and other graduate students, influencing departmental policies.

References

Dr. Jason Dexter, University of Colorado Boulder—jason.dexter@colorado.edu
Dr. Keith Horne, University of St Andrews—kdh1@st-andrews.ac.uk
Dr. Erica Nelson, University of Colorado Boulder—erica.june.nelson@colorado.edu
Dr. Ric Davies, Max Planck Institute for Extraterrestrial Physics—davies@mpe.mpg.de